

D424 – Software Engineering

Task 3



Capstone Proposal Project Name: _____ Degree Planner

Student Name: _____ Galen Hammerle

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Task 3 Design Document

Application Design and Testing

Class Design

The project utilizes two main elements: a user interface and a database. UML details of each design are provided below.

Database Repository Design

An interface defining the API to the database allows the database to be abstracted from the user interface. The `DataRepositoryBase` class is abstract and implements `IDataRepository`. `DataRepositoryBase` can implement methods that are not reliant on the concrete database implementation. This pattern allows us to keep business logic out of the user interface. This pattern also makes implementing a different database easier if it needs to be changed.

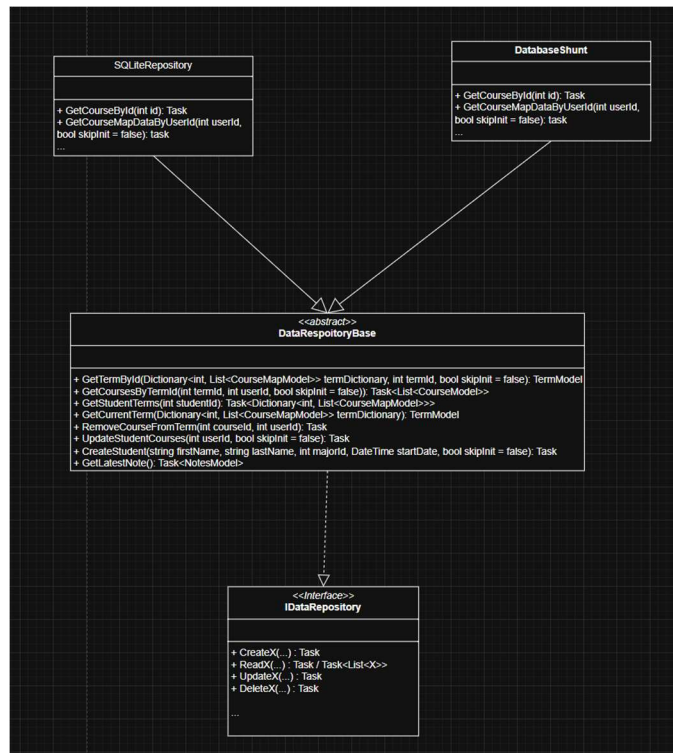


Figure 1 Data Repository UML

User Interface Design

BaseContentPage was created to provide standard functionality to all pages of the application. This reduces duplicated code and standardizes functionality. This base class provides:

- The ability to present dialogs to the user for success and errors.
- Shared access to IDataRepository and ActiveUserId.
- Event logging functionality.
- Report generation.

Reports generated from a page provide a ContentView to the ReportPage (a ContentPage). This allows for easy addition of reports; create a new ContentView for the report and update the page to call the inherited ShowReportPopupAsync method.

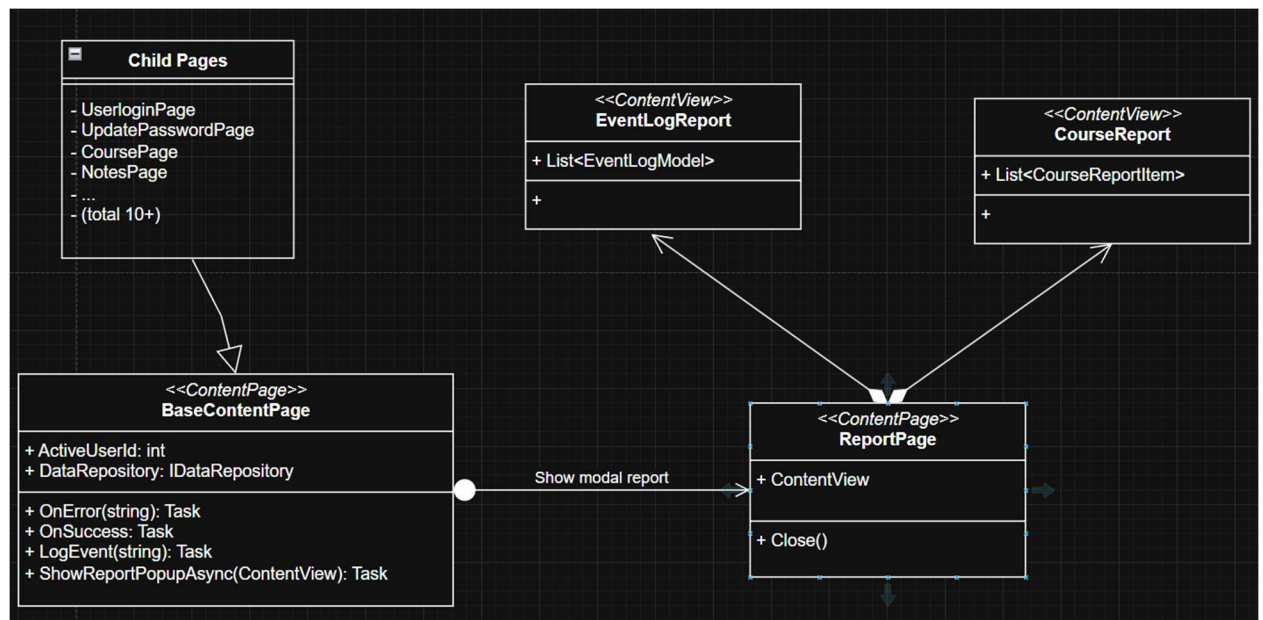


Figure 2 Content Page UML

UI Design

The user interface was designed to support two types of users, administrators and students. The user interface was mocked using a low-fidelity wireframe created with Software Ideas Modeler. Students are presented with the current or following term when they log in. Students can then view courses in the term and add notes for each course. Administrators are presented with a list of tasks they can perform. The low-fidelity wireframe captured in the images below served as a guide to the application's development.

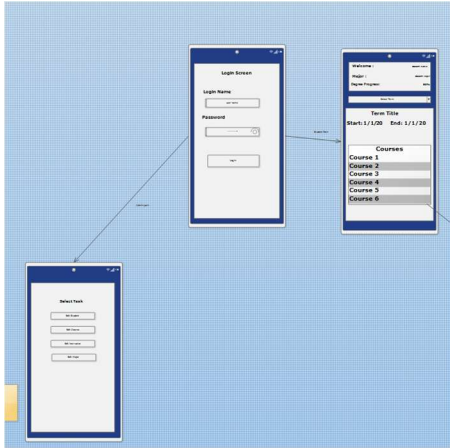


Figure 3 Initial Flow

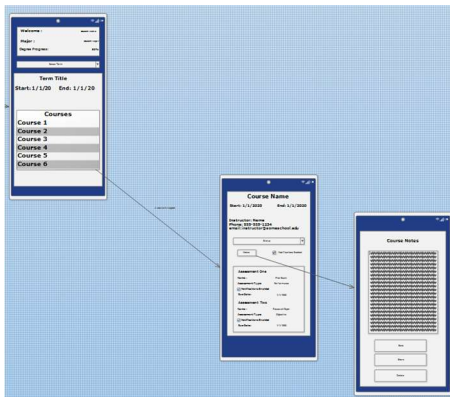


Figure 4 Student Workflow

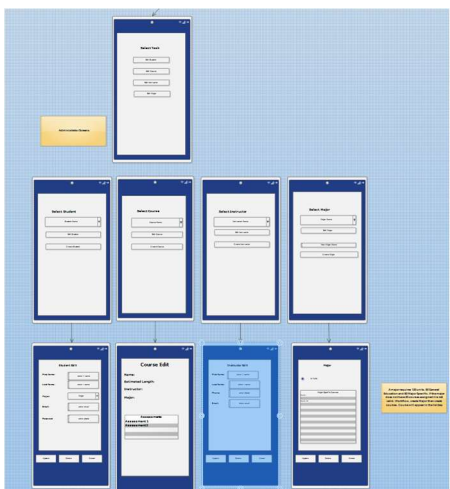


Figure 5 Admin Workflow

Unit Test Plan

Introduction

Purpose

Most of the testing of this application was functional testing performed during development. Unit tests were also utilized to test portions of the code. The testing used ensured the reliability and functionality of the application.

Overview

The application is delivered with files to seed the database. These files are installed with the application. The database will be seeded when a user attempts to log in for the first time. The database will be seeded with three users: one administrator and two students. With the seeded data, all functional testing is possible. During the development phase, the functional tests were performed. Errors encountered during development were resolved using the Visual Studio 2022 built-in debugger, adding code to write to the console, and conducting internet searches.

The following section details functional tests for key features of the two user types and unit tests.

Test Plan

Items

The latest version of the application source code is located here: <https://gitlab.com/wgu-gitlab-environment/student-repos/ghamme3/d424-software-engineering-capstone.git>

Visual Studio 2022 Community Edition, with support for .NET MAUI development.

Features***General Features***

- User security and login.
- Database seeding

Administrator Features

- Create a new major
- Create new courses in General Education & for the new major
- Create and Delete a student
- View Reports for courses, the event log

Student Features

- View terms
- Drop a course, major & general education
- Add a course, including search
- Add notes for a course

Deliverables

There are no expected deliverables for the functional test being performed. Any testing that did not result in a working feature or function was addressed immediately. Once a key feature was considered working, the code would be committed to the branch. If the testing was performed by an independent user, an issue detailing the concern should be created in the GitLab repository.

Unit tests are expected to be run and all pass prior to pushing any changes to the repository.

Set up for Functional Test Session using Visual Studio

Open the solution using Visual Studio 2022 Community Edition.

Rebuild the solution.

Start the emulator.

Tools → Android Device Manager

Start the device (Pixel 7 – API 35 with Android 15.0 – API 35 used for all testing and development)

Clean old versions and data as necessary. (New installations will not require this step)

Tools → Android → Android Adb Command Prompt

Run this command *adb shell pm clear com.companyname.c424assessment*

The command will return 'Success'. If no devices are found, make sure the device is running.

Start the application without Ctrl+F5

Set up for Functional Test Session using an Android device

..... install the application.....

..... start the application

Application Start

When the application has been successfully installed and started, a dialog asking to allow notifications will appear. Click 'Allow'.

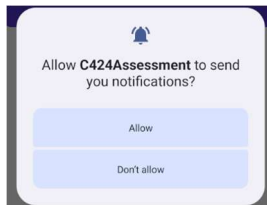


Figure 6 Notification permission

The user login screen will appear. This is the entry point for all users.

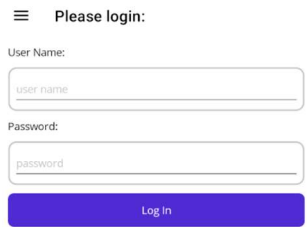
A screenshot of a user login screen. At the top left is a hamburger menu icon. To its right is the text "Please login:". Below this, there are two input fields. The first is labeled "User Name:" and contains the placeholder text "user name". The second is labeled "Password:" and contains the placeholder text "password". Below the password field is a blue button with the text "Log In" in white.

Figure 7 User login Screen

The application is now ready for functional testing.

User Security and Login Test

The first attempt at a login will seed the database (a slight lag is possible) with three users and passwords.

- User Name: admin Password: Admin123!
- User Name: Beth.Santos Password: BethSantosPW
- User Name: Jake.Smith Password: JakeSmithPW

Pass/Fail Criteria

If an incorrect user name or password is entered, the user will be prompted to resolve the issue.

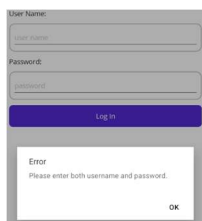
A screenshot of the login screen showing an error message. The "User Name:" and "Password:" fields are visible. Below the "Log In" button, there is a white box with a red border containing the text "Error" and "Please enter both username and password." with an "OK" button at the bottom right.

Figure 8 Invalid Login Example

Upon a successful login user `admin` will navigate to the Select Task Page.

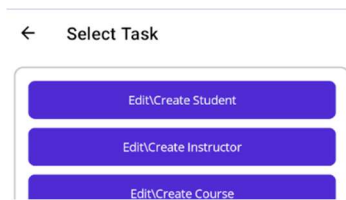


Figure 9 Select Task Page

Users 'Beth Santos' and 'Jake Smith' will be asked to change their passwords.

Figure 10 Update password Page

Password updates are validated against the correct current password, and the new password entries must match. (For simplicity, there are no added requirements for the new password.) The user will be prompted for validation issues or success. After a successful change, the user will return to the login page.

Database Seeding Test

The act of logging in is expected to seed the database. This can be validated by logging in as 'admin' and viewing the Course Report.

Log in using 'admin' password 'Admin123!'.

Click the Generate Course Report button.

Course Report shows a list of all courses, the associated major (if any), and the number of students enrolled.

Intro to Computer Literacy	General Studies 1
Introduction to Astronomy	General Studies 0
Human Growth and Development	General Studies 2
Introduction to Logic	General Studies 1
Writing Fundamentals	General Studies 1
Intro to Economics	General Studies 1

Figure 11 Course Report Snip

Pass/Fail Criteria

The database is expected to be seeded with a list of courses, two students and two majors. The two students are enrolled in different degree plans, one for Computer Science and the other for Biology. The Course Report will reflect this, as the course for each major will have one student enrolled.

Human-Computer Interaction	Computer Science 1
Information Retrieval	Computer Science 1
Natural Language Processing	Computer Science 1
Theory of Computation	Computer Science 1
Cell Biology	Biology 1
Molecular Genetics	Biology 1

Figure 12 Major Enrolled Students Snip

Create and Delete Student Test

The admin user is expected to have the ability to create, update, and delete users. Users who have started courses cannot be deleted.

Log in using `admin` password `Admin123!`.

Select Edit\Create Student

Complete the fields as in the figure below. And click Update.

Figure 13 New Student Creation

Return to the Create Student Page and select the new student. The data should be populated with what was entered. Return to the Select Task page using Cancel.

Generate a course report and notice that some General Studies courses now have 3 enrolled and the major listed above now has 2 students enrolled.

Close the report and return to the Edit Student page.

Select and delete the new student.

Verify all student data has been removed by viewing the course report.

New students will be named First.Last and have a default password of FirstLastPW.

Pass/Fail Criteria

The Select Student Picker is expected to list all students.

Data entry fields shall be validated.

The Select Major Picker is expected to list all active majors.

(An active major is one with 20 related courses.)

If a student with progress toward their degree is selected, the data field will populate, and the user cannot be deleted.

The Reset Password check box shall be enabled.

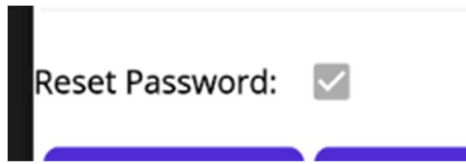


Figure 14 Failed Result Reset Password

The Start Date Picker will not allow selection of dates in the past.

The Start Date Picker will be disabled for students who have started their degree plan.

For a new student Reset Password value will not be honored when the student is created.

If the new student is deleted, all mapped course data will be removed.

Edit Create a Major Test

The admin user can create Majors and courses. A major must have 20 course to be considered active. Courses cannot be shared between majors. Inactive majors can not be assigned to students.

Log in using `admin` password `Admin123!`.

Select the Create/Edit Major Button.

Select an existing major and observe behavior.

Cancel and return to the Create/Edit Major page.

Add a name for the new major, click Update.

Return to the Create/Edit Major page.

Select the recently added major. Click add course.

The course edit page will appear, allowing the user to enter course information.

←

Course Name: New Course One

New Major

Instructor

David Martinez

Instructor: David Martinez
Phone: 458-555-1204
Email: David.Martinez@this.edu

Course Duration: 3

Assessment One Final Exam Performance

Assessment Two Assessment Two Name Select Type

Update Delete Cancel

Figure 15 Course Edit Page

Click update. The Create/Edit Major Page will reload with the created course.

Return to the Select Task page using Cancel.

Go to Edit/Create Student page. Try to assign the new major. The new major will not appear until it is active (has 20 courses).

View the Course Report and see that the new course and major are in the database.

Capstone Project in Biology	Biology	1
New Course One	New Major	0

Figure 16 New major and Course Report

Once all 20 courses are assigned to the major, a new student can be enrolled for the degree plan.

Return to Create/Edit major and delete the major.

Pass/Fail Criteria

Data and action validation issues will notify the user.

Any major in which a student is enrolled cannot be deleted.

A major cannot have more than 20 courses.

A course created for the major will be linked to the major.

The Edit\Create course page will validate inputs.

Course name

At least one assessment.

If there are two assessments, they must be of different types.

Student Workflow Test

Students will have the following abilities:

- View all terms
- Remove a course from a term.
- Add a course to a term.

Log in as Beth.Santos password BethSantosPW unless it was updated.

(If the password was forgotten, use the admin account to reset the password to the default, or reseed the database as admin.)

In the current term click on a course to view the details.

Change the status of two courses: set one to Active and one to Completed.

For a third course, disable notifications

Use the update button to save the changes.

Log out and back in as the same user.

A different set of notifications should occur.

Select a course to remove from the term.

Pass/Fail Criteria

User Beth.Santos and Jake.Smith have seeded data as if they have started their respective degree programs.

Notification for seeded users will generate notifications. If the course has not started the notification will inform the user that the term is ending soon.

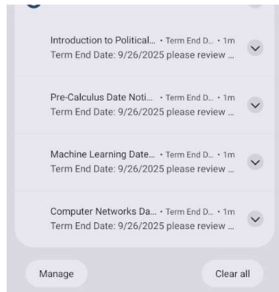


Figure 1717 Term Ending Notification

Completed courses and courses with notifications disabled will not have any alerts. An active course will notify the user of available assessments.

Completed courses cannot be removed from the term.

Setting a course status to Completed will update the percentage complete value.

A course must have at least one course and no more than six.

If a course is added from a different term. Both terms will reflect this change.

Major courses cannot be dropped; they will be moved to the following term, which can hold the course.

If a General Studies course is dropped, the student will see that they are not enrolled in enough units for graduation.

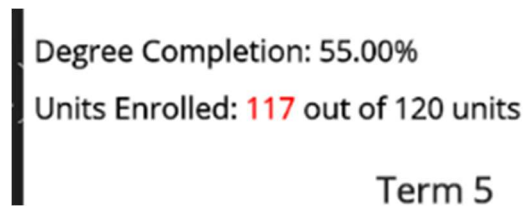


Figure 188 Not Enough Units

Prior to dropping a course, the student will be notified of the expected result of the action and the option to continue.

Unit Tests

The repository contains unit tests. These tests can be run using Visual Studio 2022 Community Edition. A shunt class derived from `DataRepositoryBase` was created to test the logic of a certain method, without relying on a concrete SQLite-PCL database to be instantiated.

The shunt class creates mock data when a test is run and provides this data to the test method.

```

192 public override Task<List<CourseMapModel>> GetCourseMaps(bool skipInit = false)
193 {
194     // return some mock data
195     List<CourseMapModel> courseMaps = new List<CourseMapModel>
196     {
197         new CourseMapModel
198         {
199             UserId = 1,
200             TermId = 1,
201             CourseId = 1,
202             CourseStatusId = 2,
203             StartDate = DateTime.Now.AddDays(-30),
204             EndDate = DateTime.Now.AddDays(30),
205             NotificationsEnabled = true,
206             NotesId = 800,
207         },
208         new CourseMapModel
209         {
210             UserId = 1,
211             TermId = 1,
212             CourseId = 2

```

Figure 19 DataRepository Shunt Mock Data

When tests are run, a common setup method creates a `DatabaseShunt`. See figure 20 for more information.

```

public class Tests
{
    public IRepositoryBase<DataRepository> DataRepository { get; private set; }

    [SetUp]
    public void Setup()
    {
        // This is a shunt for the IRepositoryBase interface
        // It will not perform any actual database operations
        DataRepository = new DatabaseShunt();
    }

    [Test]
    public async Task GetStudentTermsTests()
    {
        var sut = DataRepository;

        var terms = await sut.GetStudentTerms(1);
        Assert.IsNotNull(terms);
        Assert.That(terms.Count == 2);
    }

    [Test]
    public async Task GetCurrentTermTest()
    {
        var sut = DataRepository;
        var courseMap = await sut.GetCourseMapDataByUserId(2);
        // termId will contain multiple entries in the map we need only unique ids
        var termDictionary = courseMap.GroupBy(t => t.TermId)
            .ToDictionary(g => g.Key, g => g.ToList());
        var currentTerm = sut.GetCurrentTerm(termDictionary);
        Assert.IsNotNull(currentTerm);
        Assert.That(currentTerm.TermId == 2);
    }

    [Test]
    public async Task GetCourseForTerm()
    {
        var sut = DataRepository;
        var courses = await sut.GetCoursesByTermId(1, 1);
        Assert.IsNotNull(courses);
        Assert.That(courses.Count == 3);
    }
}

```

Figure 20 Test Case Examples

Visual Studio discovers unit tests when the project is built. To run the tests, use the Test Explorer window or, from the navigation menu, select Test → Run All Tests.

All Results from the unit test are visible in the Test Explorer window in Visual Studio.

Test	Duration	T
▲ ✓ DataRepositoryTests (3)	8.8 sec	
▲ ✓ DataRepositoryTests (3)	8.8 sec	
▲ ✓ Tests (3)	8.8 sec	
✓ GetCourseForTerm	8 ms	
✓ GetCurrentTermTest	8.8 sec	
✓ GetStudentTermsTests	54 ms	

Figure 21 Unit Test Results

User Guide

Introduction

This user manual will provide installation instructions and key workflow steps for the two user types. Instructions for the admin user to log in and create a new major, including creating students and courses. The instructions for student users will cover the primary student workflow of viewing a term, viewing a course in the term, updating course status, disabling notifications, and adding notes to a course. Directions on dropping and adding a course will also be provided.

Installation and Using the Application

The application apk file is available for download on Github:

<https://ghammerle.github.io/WGU-D424-Capstone/>.

The source code is available on Gitlab:

[Task 3 · working · WGU GitLab Environment / Student Repos / ghamme3 / D424 Software Engineering Capstone · GitLab](#).

Installation (Visual Studio 2022 Setup)

All instructions below assume the computer is set up for mobile .NET MAUI development using Visual Studio 2022 Community Edition, git is installed, and the computer has internet access.

PC Setup for development and testing:

- Windows 11
- Intel i5-12400 12th Gen
- Radeon RX550/550 Series GPU
- 16GB of RAM

(Testing conducted on a PC with lesser graphics performance resulted in emulator launch issues. The emulator had to be started through the command line to avoid the graphics-related issue. “C:\Program Files (x86)\Android\android-sdk\emulator.exe” -avd pixel_7 -_api_52_1 -gpu swiftshader_indirect -verbose, it is best to avoid this complication.)

Open the Command Prompt. (Type cmd, and press Enter in the Windows search bar)

Create a working directory.

Navigate to the working directory and type git init to initialize git for this folder.

Navigate to the source code repository:

[Task 3 · working · WGU GitLab Environment / Student Repos / ghamme3 / D424 Software Engineering Capstone ·](#)

[GitLab](#)

Clone the repository, click code, and copy the path based on your account configuration.

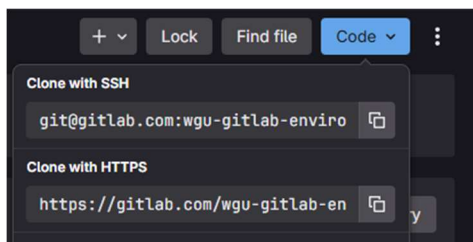


Figure 22 Git clone path

Return to Command Prompt, type git <path> (paste the copied path using Ctrl + V), and press Enter.

The repository will be cloned to the directory.

Check out the branch named working.

git checkout working

Open Visual Studio, open the solution C424 Capstone.sln

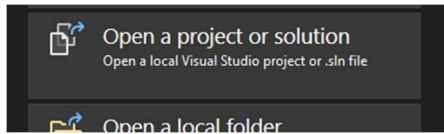


Figure 23 Open Solution

A directory navigation window will appear when the above selection is made.

Once the solution is loaded, use Tool → Android → Android Device Manager

Set up an Android device, the default was used to generate:

Name	OS	Processor	Memory	Resolution	Action
 Pixel 7 – API 35 + Google Play	Android 15.0 – API 35	x86_64	1 GB	1080 x 2400 420 dpi	 Stop

Figure 24 Android Pixel 7

Login

The application database comes seeded with three users.

- User Name: admin Password: Admin123!
- User Name: Beth.Santos Password: BethSantosPW
- User Name: Jake.Smith Password: JakeSmithPW

Students added to the database will have login credentials of

- User Name: First.Last Password: FirstLastPW

Students will be prompted to update the password on their initial login or if the admin resets it to the default password.

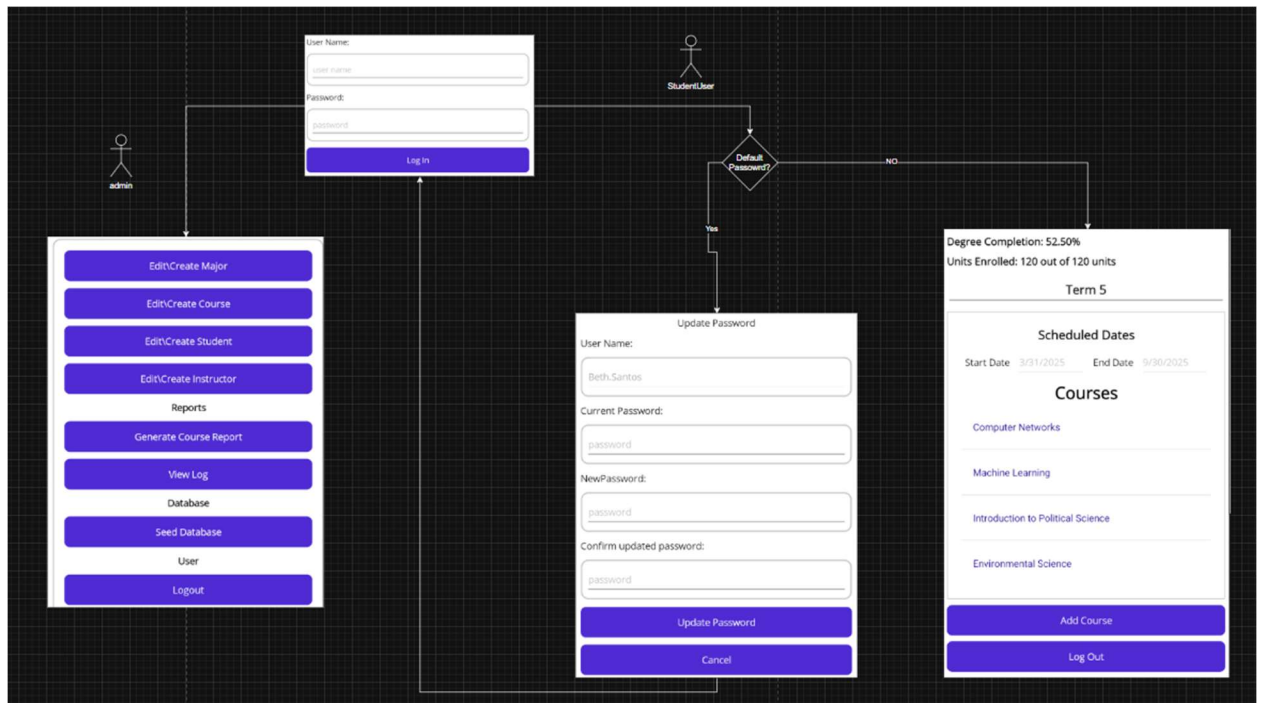


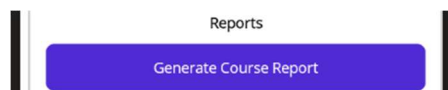
Figure 25 User Log in Flow

Administrator Workflow

The workflow detailed below will touch on major features available to the administrator user. Items covered will be adding and updating courses, majors, students, and instructors.

Viewing reports and database seeding will also be covered.

1. *Log in as admin.*
 - a. User Name: admin Password: Admin123!
2. *On the Select Task page, select Generate Course Report*



- a. *This will show a report of all courses in the database, the associated major, and the number of students enrolled. Notice: Test Major has zero students enrolled.*

b. Close the report

Test Major Course 15	Test Major	0
Test Major Course 16	Test Major	0
Test Major Course 17	Test Major	0
Test Major Course 18	Test Major	0

Close

3. Select Edit/Create Student



a. Using the Select Student Picker select a student.

i. Students who have started courses can only have their passwords reset to default.

ii. Cancel and return to the Edit Student page.

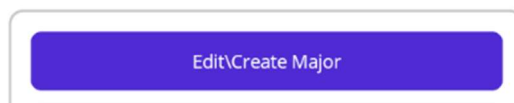
b. Using the select Major picker choose a major

Major:
Select Major

i. Notice only active majors are available. An active major is one with 20 associated courses.

c. Close the Edit Student Page using the Cancel button.

4. From the Select Task page select Edit\Create Major.



a. Select a Major

b. The associated course will appear.

Courses	
Digital Media Literacy	
Data Structures and Algorithms	

c. Tap (click on) a course

Course Name: Data Structures and Algorithms

Select Major

Instructor
Lily Robinson

Instructor: Lily Robinson
Phone: 458-555-1217
Email: Lily.Robinson@this.edu

Course Duration: 2.5

Assessment One
Final Exam Objective

Assessment Two
Algorithm Implementation Project Performance

Update/Save Delete Cancel

i. If a course is assigned to a student only the instructor can be edited.

d. Click Cancel to return to the Edit Major page.

e. Click the Add Course Button.

Add Course

i. If the Major already has 20 courses, you cannot add a course.

f. From the Create/Edit Major page use the Major Picker to select Test Major.

i. This major was seeded with 18 courses for demonstration and training purposes.

g. Click Add Course

h. Enter a Course Name, select an instructor, and add at least one assessment.

Course Name: My New Course

Select Major

Instructor
Michael Davis

Instructor: Michael Davis
Phone: 458-555-1202
Email: Michael.Davis@this.edu

Course Duration: 2.5

Assessment One
Final Exam Performance

Assessment Two
Final Essay Objective

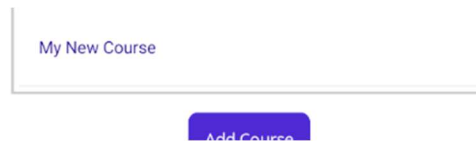
Update/Save Delete Cancel

i. *A course must have one assessment and no more than two.*

Assessments cannot be of the same type (Objective & Performance).

ii. *Click Update\Save to add the course to the major.*

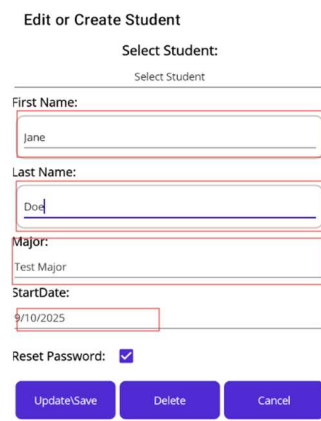
iii. *The course will appear in the list for the selected major.*



i. *Add a second course to Test Major; making it active.*

5. *Return to the Edit\Create student page.*

a. *Create a student and assign the major of Test Major. Provide first and last names and a start date. (Start Date can no be in the past.)*



b. *Click Update\Save to add the student.*

6. *To verify if a student was added to the major use the Generate Course Report button to view the course report.*

a. *Notice that Test Major courses have one student enrolled and some general education courses now have three students enrolled.*

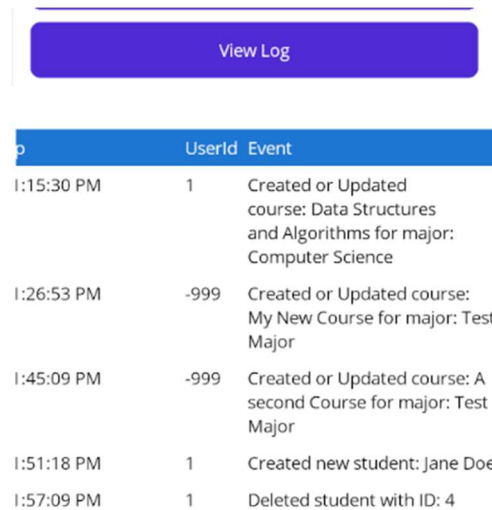
b. *The two courses added earlier will also appear in the report.*

7. *Delete a Major.*

- a. *Open the Create\Edit Major, select any major*
 - i. *Majors which have students enrolled in cannot be deleted.*
- b. *Return to the Task Selection page using Cancel.*

8. *Delete a student*

- a. *Open the Create\Edit Student page, select the newly added student.*
 - i. *Students who started courses cannot be deleted.*
- b. *Click the delete button to remove the selected student.*
- c. *Validate the student has been deleted using the Course Report.*
 - i. *The View Log button on the Select Task page will generate a report of the event log showing various events.*



The screenshot shows a blue button labeled 'View Log' at the top. Below it is a table with three columns: 'Time', 'UserId', and 'Event'. The table contains five rows of log entries.

Time	UserId	Event
1:15:30 PM	1	Created or Updated course: Data Structures and Algorithms for major: Computer Science
1:26:53 PM	-999	Created or Updated course: My New Course for major: Test Major
1:45:09 PM	-999	Created or Updated course: A second Course for major: Test Major
1:51:18 PM	1	Created new student: Jane Doe
1:57:09 PM	1	Deleted student with ID: 4

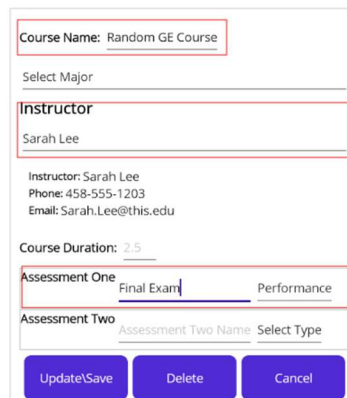
- d. *Open Create\Edit Major*
 - i. *Select Test Major from the Select Major picker.*
 - ii. *Click Delete, if no students are enrolled in this major the major can be deleted.*
- e. *Return to the Select Task Page and select Edit\Create Course.*

- i. *If a major does not contain 20 courses, it will be available from the Select Major picker.*



- ii. *Create a new course for General Education*

1. *Like adding course to a major directly, fill in the data but do not select a Major. Save the changes.*



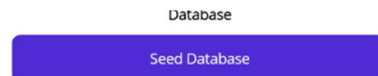
2. *Use Generate Course Report to validate the course was saved.*



Random GE Course	General Studies: 0
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- f. *Seeding the database*

- i. *The admin user can reset the database to the initial values using the Seed Database button on the Select Task page.*



- ii. *Click the button and wait for the dialog informing you that the database action is completed.*

Student Workflow

1. Log in as a student. (Beth Santos used for the example)

- a. There are two students provided.
 - i. Beth.Santos Password BethSantosPW
 - ii. Jake.Smith Passoword Jake.SmithPW
- b. Newly created students are defined with a pattern
 - i. FirstName.LastName Password : FirstNameLastNamePW
- c. When the password is the default students will be prompted to enter a new password.

Update Password

User Name:

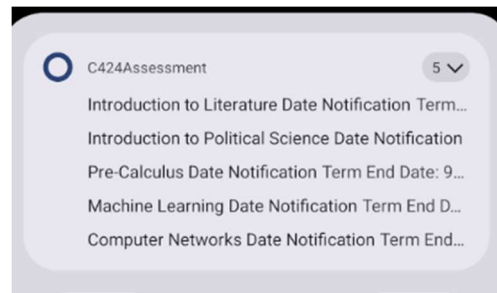
Current Password:

NewPassword:

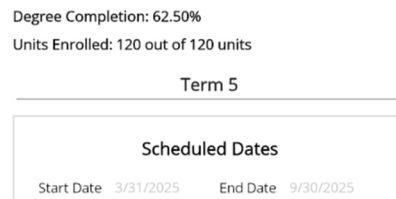
Confirm updated password:

- i. The current password must be entered, and the new password must be entered twice for confirmation. *(There is no secondary validation of passwords for length or reuse. This is done to simplify rapid testing.)*
- ii. After updating their password, users will need to log in using their new password.

- iii. The students provided were set up as if they were nearing the end of a term. When logged in, they should receive notifications.



- iv. Students will see their progress and Units enrolled. The application will open to their current term.



- d. Drop a course from a term.

- i. Select a Course

1. A course can be dropped from a term as long as the term will have at least 1 course remaining. If a general course is dropped, the user will need to add a course to get back to the required 120 units to graduate. Courses required for the enrolled major will move to the next available term.
2. Pop-ups will occur to let the user understand the action.

Drop Course

This course is part of your degree plan. You will need to add another course to reach 120 units. Are you sure you want to drop it?

No Yes

Drop Course

This course is part of your major and will be moved to a later term.

Cancel Continue

3. Courses are randomly selected so that the examples may vary.

ii. When a course is selected, the course view will open

Course Editor

Introduction to Astronomy

Starting Date: 3/31/2025 Ending Date: 9/30/2025

Status: Planned Notifications: ☒

Instructor: David Martinez
Phone: 458-555-1204
Email: David.Martinez@this.edu

Notes

Performance Assessment

Star Chart Project Notifications: ☒
Due Date: 09/30/2025

Objective Assessment

Final Objective Exam Notifications: ☒
Due Date: 09/30/2025

Update

Remove Course

Return to Term

iii. Select Remove Course

1. After removing one general education course and one major course from the term, the user will see fewer courses, and the enrolled units will be updated.

Degree Completion: 55.00%
Units Enrolled: 117 out of 120 units

Term 5

Scheduled Dates

Start Date 3/31/2025 End Date 9/30/2025

Courses

Computer Networks

Introduction to Literature

Pre-Calculus

iv. Select Add Course

1. The user can select a course to add to the term.
2. The Select a Course picker will list courses available to the user.
 - a. This includes courses that may exist in future terms.
3. A search filter is available to filter the selection.

Course Selection Page

Search Selectable Courses:

Sci

Select a course.

Course to Add:

Select a course.

Physical Science

Environmental Science

Introduction to Political Science

Cancel

- v. Click Add Course once a selection has been made.

- vi. The course will be added to the current Term.

Degree Completion: 55.00%
Units Enrolled: 120 out of 120 units

Term 5

Scheduled Dates
Start Date: 3/31/2025 End Date: 9/30/2025

Courses

Computer Networks

Introduction to Literature

Pre-Calculus

Introduction to Political Science

Add Course

Log Out

- e. Updating Notifications

- i. Notifications are for a course and will be triggered if a course has not started and the term will end in the next 30 days.
- ii. Notifications can be disabled by course and assessment. Course settings take precedence.

Status: Planned Notifications: ☒

Instructor: Andrew Harris
Phone: 458-555-1212
Email: Andrew.Harris@this.edu

Notes

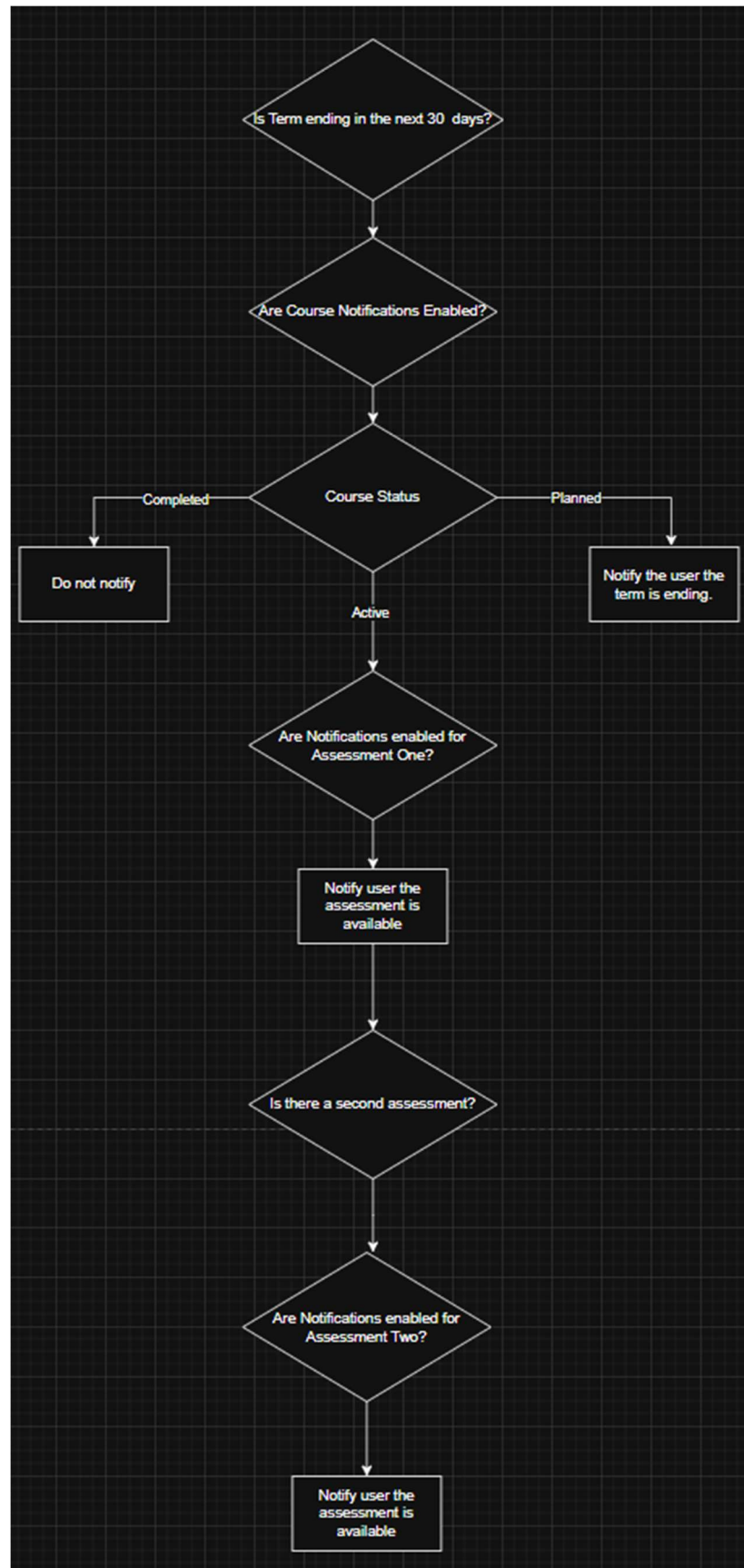
Performance Assessment

Policy Review Notifications: ☒
Due Date: 09/30/2025

Objective Assessment

Comparative Politics Quiz Notifications: ☒
Due Date: 09/30/2025

- iii. Notification flow chart



f. Updating Course Status

- i. Navigate to the current term
- ii. Use the status picker to set a course to completed.

Introduction to Political Science

Starting Date: 3/31/2025 Ending Date: 9/30/2025

Status: Completed Notifications: ☒

Instructor: Andrew Harris

- iii. Click Update to save the changes.
- iv. Notice that the degree completion percentage has been updated.

Degree Completion: 57.50%

Units Enrolled: 120 out of 120 units

g. Adding and sharing course notes.

- i. From the Term Page, select a course.
- ii. In the Course Page, click the Notes button.

Email: Andrew.Harris@this.edu

Notes

- iii. Add content in the note entry field.

Notes for Introduction to Political Scien...

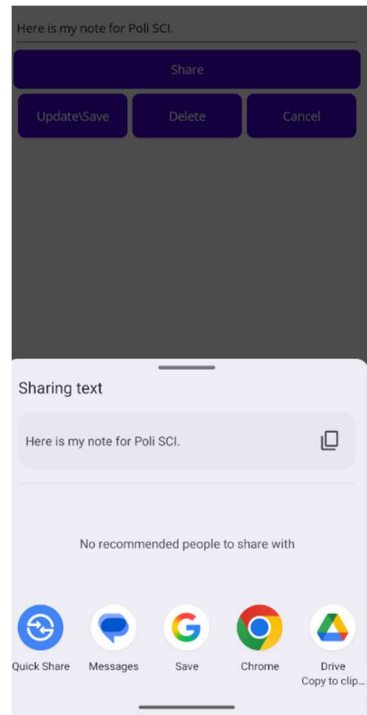
Enter your note here.

Share

Update/Save Delete Cancel

- iv. Click Update/Save
- v. Return to the recently created note.

- vi. Use the Share button to bring up the sharing window.



Git Logs

The git logs generated for this project are contained in the images below.

Commits to the D424 Capstone repository in GitLab.

WGU GitLab Environment / Student Repos / ghamme3 / D424 Software Engineering Capstone · GitLab

Uncommitted changes	6 Sep 2025 17:14
Task3 Word document user manual, fixed issue related to Term dates when moving a course	6 Sep 2025 17:14 GHammerle <875> 2c764b2
add test data and addresses minor navigation issues	6 Sep 2025 15:07 GHammerle <8750> 3c01888
origin/working resolves issue with invalid seeding data and assessment being saved to the database, resolves application crashes related to the edit major page, adds ability to delete unmapped courses	1 Sep 2025 17:56 GHammerle <8750> 43d7404
resolves the issue where a deleted student's mapped course data is not deleted, also cleans up Course page UI	1 Sep 2025 14:22 GHammerle <8750> 686d261
update task 3 documentation, resolves build issue and user password update issue found during testing	1 Sep 2025 13:50 GHammerle <8750> 9a6f64c
updates notifications, add improvements to support rotation of the device, clears login screen data entry fields	31 Aug 2025 12:36 GHammerle <8750> 71d5943
adds ability for user to add a course, updates seeding to use old dates, fixed percentage issue	30 Aug 2025 18:52 GHammerle <8750> b20926e
add updates to Student Course view, correct term dates, notes working per student, assessment view cleaned up	30 Aug 2025 15:12 GHammerle <8750> 2c48d2b
missed files for unit tests, implemented navigation to course page from term course list	28 Aug 2025 0:28 GHammerle <8750> d2cafe1
courses are now showing in the term view when a user logs in	24 Aug 2025 18:41 GHammerle <8750> 4ed97f1
remove duplicated databasecode, termmodel now loads when student logs in	24 Aug 2025 14:22 GHammerle <8750> c104d86
add unit test for DB, begin refactor to separate data layer	24 Aug 2025 11:32 GHammerle <8750> 90a4a70
reuse new update course method	23 Aug 2025 21:03 GHammerle <8750> 27bce15
new student course map now populates	23 Aug 2025 20:45 GHammerle <8750> 9ab-d0f1
adds CreateMajorPage, updates Course page to show data	23 Aug 2025 13:11 GHammerle <8750> 85e2c84
updated db seeding, edit instructor page navigation, added startdate to student model edit student page	22 Aug 2025 0:04 GHammerle <8750> c2116ae
add Log Event Report	21 Aug 2025 0:28 GHammerle <8750> e9b6396
login page update password create student pages working	20 Aug 2025 0:43 GHammerle <8750> 0d03a11
extra files to commit	17 Aug 2025 23:41 GHammerle <8750> b3b76c1
added report for all courses, updated Course db model, added admin button to seed database	17 Aug 2025 23:39 GHammerle <8750> 654c371
abstract database repository, add edit student page	17 Aug 2025 12:28 GHammerle <8750> 4cxa3e4
add user login and password update pages	16 Aug 2025 21:09 GHammerle <8750> 1d0177b
add user login page, password support, seed User db Table	16 Aug 2025 15:40 GHammerle <8750> bb58c4d
add new dbmodels and rename project	11 Aug 2025 20:32 GHammerle <8750> 6607c85
refactor database seeding, update project documents	10 Aug 2025 12:10 GHammerle <8750> a730778
moved to 424 from C971	3 Aug 2025 20:48 GHammerle <8750> 63b0dfe
add task3 files, updated with submission of task 2	3 Aug 2025 20:24 GHammerle <8750> f4f99c7
Initial commit	27 Jul 2025 20:49 GHammerle <8750> 4ea53a8
origin/main origin/Working Corrected misspelling	16 Jul 2024 19:48 Emilio Miller <emi> 9c6dfcc
Added Branching instructions to README	27 Feb 2023 10:18 Zohar Sajith <zoha> 808c354
Added WGU Template README.md	13 Feb 2023 7:45 Zohar Sajith <zoha> 5c5cab3
Initial commit	13 Feb 2023 7:42 Zohar Sajith <zoha> 835201f

Commits that created the code “Moved from C971”

WGU GitLab Environment / Student Repos / ghamme3 / C971 Mobile Application Development Using C Sharp · GitLab

Uncommitted changes	6 Sep 2025 17:01
update project word doc	22 Jul 2025 22:07 GHammerle <875> b691605
Task C4 update courses were not deleted from the database, added DeleteCourse Method and call	22 Jul 2025 21:18 GHammerle <8750> 4d3ac1b
Per Task C4 assessment validation fixed not allowing assessment of the same type	22 Jul 2025 21:02 GHammerle <8750> dea4e46
fix deadlock during seeding, reading assessments, Add document related to Task D1	20 Jul 2025 16:42 GHammerle <8750> 142841e
fix bug saving course data with one assessment testing for C4	20 Jul 2025 15:56 GHammerle <8750> 5e79b25
Fix new course validation per Task C3	20 Jul 2025 14:37 GHammerle <8750> 7243114
fixes term validation bugs	20 Jul 2025 11:35 GHammerle <8750> 98262f2
add ability to toggle course notifications	20 Jul 2025 10:59 GHammerle <8750> e90bc44
creates NotificationHelper, Called from App.xaml, resolve threading and deadlock issues	20 Jul 2025 10:18 GHammerle <8750> 869682c
update to support Notifications for assessments and courses	19 Jul 2025 10:54 GHammerle <8750> d434d1d
add course edit page, add sharing to Notes, other clean up	15 Jul 2025 21:01 GHammerle <8750> 12ec909
remove OnTapped Event from Instructor View, selector allows user to change instructor	14 Jul 2025 11:29 GHammerle <8750> d02d3c3
Added new button to add Course Term and Instructors. Some bugs left to sort out	14 Jul 2025 0:49 GHammerle <8750> 28658ae
Add page for Term&Edit CRUD features for TermModel is functional including validation	13 Jul 2025 14:13 GHammerle <8750> 9b23641
instructor CRUD is functional	12 Jul 2025 18:09 GHammerle <8750> 89c792f
Notes CRUD functionality working	12 Jul 2025 16:17 GHammerle <8750> 1f8c90e
database seeding working, term view opens, course view opens	12 Jul 2025 14:47 GHammerle <8750> 3c52fed
add remaining tables, remove flat file persistence, add CRUD methods for tables, Builds, error ignoring Notifications	12 Jul 2025 1:54 GHammerle <8750> 688a3da
working thru assessment model changes causing crashes	11 Jul 2025 0:46 GHammerle <8750> 9056d20
add assessment table to database	9 Jul 2025 8:30 GHammerle <8750> 0c73997
add attributes for SQLite to data models	7 Jul 2025 22:17 GHammerle <8750> 127b2ee
update term view to have finish buttons, add sqlite-net-pcl reference	6 Jul 2025 20:33 GHammerle <8750> 69cc08f
add Notes View can add edit and delete notes for a course (no sharing at this time) task C1 implement B2 item 5	6 Jul 2025 14:03 GHammerle <8750> 7dc59c9
add logic to assessment edit page, adds validation to courses and instructors, data now persists	5 Jul 2025 15:13 GHammerle <8750> 912ae81
Add instructor edit page save data and validates data is not persisting with app starts task C3	4 Jul 2025 18:16 GHammerle <8750> a73272a
add buttons to edit and cancel, Cancel navigation from course to term is working	4 Jul 2025 12:55 GHammerle <8750> 9e605e6
Add wireframe PDF B1,B2,B3	2 Jul 2025 22:26 GHammerle <8750> 1a7f66e
add TermView	30 Jun 2025 23:30 GHammerle <8750> 31f5a87
move files for mock data refactor	30 Jun 2025 1:49 GHammerle <8750> 6c1a27a
refactored mock data and committing of large files	30 Jun 2025 1:03 GHammerle <8750> 122a665
add instructor to mock dataset per requirement C6	25 Jun 2025 21:11 GHammerle <8750> d11acae
clean up assessment view	22 Jun 2025 20:13 GHammerle <8750> 73ae341
assessment control added to course view, renamed many items fixing spelling error	22 Jun 2025 15:59 GHammerle <8750> 0e4b44e
update Course view, added routes, rename and move files STATE Term 0 loads course navigation works	22 Jun 2025 13:08 GHammerle <8750> a5c7d38
creates mock data	21 Jun 2025 21:08 GHammerle <8750> b520844
Create and add details to data models	21 Jun 2025 13:07 GHammerle <8750> 405569f
Add basic UI Elements	21 Jun 2025 10:58 GHammerle <8750> 4e39333
Initial commit	10 May 2025 13:48 GHammerle <8750> df354c9
main 1f1 Initial commit	10 May 2025 13:41 GHammerle <8750> b9cf1ad
origin/main origin/HEAD Initial commit	31 Jul 2023 11:25 Matt Alexander <n> 447c59d